

ECEN-662: Homework 1

Due on January 27, 2026 at 11:59 PM

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Problem 1

Let $X \sim p_\theta(x)$, where

$$p_\theta(x) = \left(\frac{\theta}{2}\right)^{|x|} (1-\theta)^{1-|x|}, \quad x \in \{-1, 0, 1\} \quad \text{and} \quad \theta \in (0, 1)$$

- i) Does $p_\theta(x)$ belong to the exponential family?
- ii) Is $|X|$ a complete sufficient statistic?
- iii) Is X complete?

Solution

Part (i): This pmf can be rewritten into a different form

$$\begin{aligned} p_\theta(x) &= \exp \left(\log \left(\left(\frac{\theta}{2} \right)^{|x|} (1-\theta)^{1-|x|} \right) \right) \\ &= \exp \left(|x| \log \frac{\theta}{2} + \log(1-\theta) - |x| \log(1-\theta) \right) \\ &= \exp \left(\underbrace{|x|}_{T(x)} \underbrace{\log \frac{\theta}{2(1-\theta)}}_{\eta(\theta)} + \log(1-\theta) \right) \end{aligned}$$

Since the model $p_\theta(x)$ can be expressed in the form $f_\theta(x) = h(x) \exp(\eta^t T - A(\theta))$, it belongs to the exponential family.

Part (ii): $T(X) = |X|$ is a complete sufficient statistic, since $\mathbb{E}_\theta[\phi(T)] = 0, \forall \theta \in (0, 1)$.

Part (iii): X is not complete since $\mathbb{E}_\theta[\phi(T)] = 0, \forall \theta \in (0, 1)$ does not imply that $\mathbb{P}_\theta(\phi(T) = 0) = 1, \forall \theta \in (0, 1)$.

Problem 2

Let $X = (X_1, \dots, X_n)$, where X_i 's are i.i.d. $\mathcal{N}(\theta, \theta)$, and $\theta > 0$ is an unknown parameter. Determine a complete sufficient statistic for θ and carefully justify your answer.

Solution

The pdf of $\mathcal{N}(\theta, \theta)$ is given by

$$\begin{aligned} f_{\theta}(x_i) &= \frac{1}{\sqrt{2\pi\theta}} \exp\left(-\frac{(x_i - \theta)^2}{2\theta}\right) \\ &= \frac{1}{\sqrt{2\pi\theta}} \exp\left(-\frac{x_i^2}{2\theta} + x_i - \frac{\theta}{2}\right) \\ &= \frac{1}{\sqrt{\pi}} \exp\left(-\frac{x_i^2}{2\theta} + x_i + \frac{\theta}{2} - \frac{1}{2}\log(2\theta)\right) \end{aligned}$$

thus $\eta(\theta) = (1, -\frac{1}{2\theta})$ and $T(x_i) = (x_i, x_i^2)$, which is a complete sufficient statistic for θ .

Problem 3

Let $X = (X_1, \dots, X_n)$, where X_i 's are i.i.d. $\mathcal{N}(\theta, \theta^2)$, and $\theta > 0$ is an unknown parameter.

- i) For $n = 1$, show that $T = X$ is a minimum sufficient statistic but is not a complete sufficient statistic.
- ii) Show that the sufficient statistic

$$T = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$$

is not complete for any $n \geq 2$.

Solution

Part (i): Here

$$\begin{aligned} f_{\theta}(x_i) &= \frac{1}{\sqrt{2\pi\theta^2}} \exp\left(-\frac{(x_i - \theta)^2}{2\theta^2}\right) \\ &= \frac{1}{\sqrt{2\pi\theta^2}} \exp\left(-\frac{x_i^2}{2\theta^2} + x_i - \frac{1}{2}\right) \end{aligned}$$

so $\eta(\theta) = (1, -\frac{1}{2\theta^2})$ and $T(x_i) = (x_i, x_i^2)$, the dimensionality difference implies that T is not a complete sufficient statistic.

Part (ii): When $T = (\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2)$, there exists a $\phi(T)$ such that $\mathbb{E}_{\theta}[\phi(T)] = 0$ but $\mathbb{P}_{\theta}(\phi(T) = 0) = 1, \forall \theta \in \Theta$, which means T is not a complete statistic when $n \geq 2$.

Problem 4

Prove the following law of total variance:

$$\text{Var}[Y] = \mathbb{E}[\text{Var}[Y|Z]] + \text{Var}[\mathbb{E}[Y|Z]]$$

for any real-valued random variable Y .

Solution

Starting from the variance definition, and applying expectation and variance properties:

$$\begin{aligned}\text{Var}[Y] &= \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 \\ &= \mathbb{E}[\mathbb{E}[Y^2|Z]] - \mathbb{E}[\mathbb{E}[Y|Z]]^2 \\ &= \mathbb{E}[\text{Var}[Y|Z] + \mathbb{E}[Y|Z]^2] - \mathbb{E}[\mathbb{E}[Y|Z]]^2 \\ &= \mathbb{E}[\text{Var}[Y|Z]] + \mathbb{E}[\mathbb{E}[Y|Z]^2] - \mathbb{E}[\mathbb{E}[Y|Z]]^2 \\ &= \mathbb{E}[\text{Var}[Y|Z]] + \text{Var}[\mathbb{E}[Y|Z]]\end{aligned}$$

Which is the law of total variance.